Phase 1 (MVP) — Work Breakdown: phases → tasks → subtasks

Phase 1 (MVP) — Work Breakdown: phases → tasks → subtasks

Revision: 2 **Last modified:** 2026-07-04T12:00:00Z **Rev 2 (hardening pass):** Added §0 cross-reference to `v07-execution/subtask-deepening-p1.md` (closes REFINEMENT_NOTES.md R5 — the PR-sized `.k` subtask breakdown for every Phase-1 task lives there, not duplicated here); added §2.1 Mermaid sequence diagram for the enroll→reach→revoke flow (ties AC1/AC2/AC6/SLO2/SLO3 to the concrete event sequence); added §2.2 phase-gate-failure / rollback protocol.

> This document is the executable spine of the HelixVPN MVP. It decomposes the Phase-1 self-hostable MVP into stable, DB-ready work items (`HVPN-P1-NNN`), each carrying *description · depends-on · deliverable · acceptance · effort · required-test-types*, grouped by control-plane module + client core/UI + platform shims + deploy + QA. It is **spec-only** — it describes *what to build and how "done" is proven*, never the product itself. Authoritative companions: the data model / DDL / protobuf live in `02-control-plane.md` (cited here as **[02]**); the data plane in `01-data-plane.md` **[01]**; the client core + UI in `03-client-core-and-ui.md` **[03]**; security / privacy / PKI in `04-security-privacy-pki.md` **[04]**; product scope in `00-product-scope-and-principles.md` **[00]**. Primary research: **[04_P1]** (`HelixVPN-Phase1-MVP.md`), **[04_P0]** (`HelixVPN-Phase0-Spike.md`), **[04_ARCH]**, **[05_YBO]**, **[SYNTHESIS]**.
>
> **PR-sized subtask breakdown (closes R5):** every task `HVPN-P1-NNN` below is decomposed one level further — into falsifiable, PR-sized subtasks `HVPN-P1-NNN.k` with their own

> acceptance/tests/complexity — in the companion `v07-execution/subtask-deepening-p1.md`. This WBS stays at task granularity (already concrete: deliverable + acceptance
>
> - tests per item); the `.k` decomposition is the R5-closing detail layer that feeds the workable-items DB (§3) at the same granularity Phase 0 uses.

Table of contents

---

0. How to read this WBS

Granularity. Three levels: **Epic** (`HVPN-P1-Enn`, a workstream) → **Task** (`HVPN-P1-NNN`, a shippable unit, ~1–8 person-days) → **Subtask** (`HVPN-P1-NNN.k`, a single PR-sized change). Every **leaf** (a task with no subtasks, or a subtask) is a workable item per §11.4.93 and is the unit that lands in the SQLite single-source-of-truth DB (§3).

ID scheme (stable, append-only, §11.4.54). `HVPN-P1-` prefix; the numeric block encodes the epic: `Enn` → tasks `nn0..nn9`, e.g. E02 (store) → `020..029`. Phase-0 entry gates use `001..006`. IDs are **never renumbered**; new work appends at the next free number in the block (gaps allowed).

Field contract (every leaf).

Field	Meaning
Desc	What the item builds, in one or two sentences (§11.4.91 ≥6 words / ≥40 chars).
Deps	Hard upstream item IDs (must be `complete` first). – = none.
Deliverable	The concrete artefact(s) — file paths, binaries, generated clients.
Acceptance	The falsifiable PASS condition, captured-evidence per §11.4.5/.69/.107.
Effort	T-shirt + person-days: XS(1) S(2–3) M(4–6) L(7–10) XL(11–15).
Tests	Required test types from the §1 closed vocabulary (§11.4.169).

Risk ordering (§11.4.132). Within and across epics, the irreversible correctness floor (RLS, no-log lint, PKI revoke, kill-switch) is sequenced first; convenience UI last.

Anti-bluff (§11.4 / §11.4.27 / §11.4.107). Every Acceptance line is a *captured-evidence* assertion, not "code compiles". A leaf is `complete` only when its required-test-types are green **with** an evidence path recorded in the item's `test_diary` (§11.4.149) and, where it has a user-visible surface, a window-scoped MP4 recording vision-verified per §11.4.159/.163. Metadata-only / config-only / grep-without-runtime PASS is forbidden.

---

1. Required-test-types vocabulary (§11.4.169)

§11.4.169 mandates that **every** workable item declare the closed set of test types it must pass before closure; the universe of types is the §11.4.27 enumeration. The closed vocabulary + canonical abbreviations used in this WBS:

Abbrev	Type	Floor expectation for HelixVPN MVP
`UNIT`	Unit	

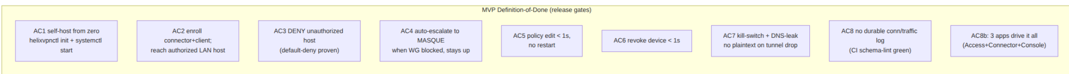
Abbrev	Type	Floor expectation for HelixVPN MVP
		pure logic (compiler, IPAM, delta diff, framing); mocks allowed **only here** (§11.4.27).
`INT`	Integration	real Postgres + Redis (testcontainers via `containers` submodule, §11.4.76); no mocks.
`E2E`	End-to-end	the netns rig [04_P0 §3] fed by the real control plane; `curl` reaches the LAN host.
`FA`	Full-automation	self-driving, re-runnable `-count=3`, no human in the loop (§11.4.98).
`SEC`	Security	RLS cross-tenant denial, key-never-leaves, mTLS, kill-switch, DNS-leak, secret-leak audit (§04).
`CHAOS`	Chaos	mid-flight SIGKILL / Redis drop / partial write; recovery restores consistent state (§11.4.85).
`STRESS`	Stress	sustained load + concurrency (≥10 parallel, ≥100 iters or ≥30 s) + boundary inputs (§11.4.85).
`PERF`	Performance	latency p50/p99 vs the SLO budget (§2); GC-tail watch on Go hot paths.
`BENCH`	Benchmarking	throughput / CPU-per-Gbps / handshakes-per-sec on the `bench.sh` rig [04_P0 §8].
`SCALE`	Scaling	N simulated agents holding streams; convergence + bounded memory over 24 h soak.
`UI`	UI	widget + golden tests on the three Flutter flavors.
`UX`	UX	flow walkthrough (enroll → connect → reach) with window-scoped MP4 + vision verdict (§11.4.159).
`REC`	Recorded-evidence	window-scoped MP4 (§11.4.154/.155) + media-validation pipeline verdict (§11.4.163).
`CHAL`	Challenge / HelixQA	a `challenges` / `helix_qa` bank entry scoring PASS only on captured evidence (§11.4.27/.107).

`DDOS` is **not-applicable** at MVP (single-node self-host, no public multi-tenant surface) — explicitly recorded `NOT_APPLICABLE: single-node-selfhost` per §11.4.6, re-armed in Phase 2 (HA / managed). Each task below lists only the **required** subset; absent types are out-of-scope for that item by design, not by omission.

---

2. MVP Definition-of-Done — the 8 acceptance gates + SLOs

The MVP ships only when **all eight** acceptance criteria (AC1–AC8) and **all four** SLOs (SLO1–SLO4) pass on a clean baseline (§11.4.40 full-suite retest, §11.4.108 runtime-signature on a fresh deploy). These are the *release gates* — restated verbatim from [04_P1 §11.2/§11.3] and bound to work-items in §21.



> AC8 in [04_P1] bundles "no durable log" with "three apps drive it"; this WBS splits the app-coverage clause out as **AC9** for traceability — both are required.

AC1 — Self-host from zero. On a fresh rootless host, `helixvpnctl init` + `systemctl --user start helixvpn-pod` yields a running control plane + edge with generated tenant/admin/CA/overlay-`/48`/creds. *Evidence:* recorded terminal session, healthchecks green, `helix_up == 1`.

AC2 — Authorized reachability. Enroll one connector advertising a LAN + one client; the client's overlay namespace `curl`s an authorized LAN host and gets the hello page. *Evidence:* E2E netns capture + pcap.

AC3 — Default-deny. The *same* client is denied a LAN host **not** granted by policy (connection refused/dropped at the edge verdict map). *Evidence:* negative E2E — packet dropped, no SYN-ACK.

AC4 — Transport auto-escalation. With plain WG UDP blocked (nft rule), the client escalates the `TransportPolicy` ladder to MASQUE/H3 and the tunnel stays up. *Evidence:* `StatusReport.transport == "masque-h3"`, tshark shows HTTP/3 (no WG signature), reachability preserved [04_P0 §5].

AC5 — Live policy reconfig. A Console/CLI policy edit reconfigures all affected devices in **< 1 s** with **zero process restarts**. *Evidence:* `helix_reconcile_seconds` p99 < 1 s histogram + Console live-event log.

AC6 — Fast revoke. Revoking a device removes its access in **< 1 s** (peer dropped from every map + kernel WG peer removed at edge + cert revoked). *Evidence:* revoke→edge-enforcement timing capture.

AC7 — Kill-switch + DNS-leak. On tunnel drop, **no plaintext egress and no DNS leak** occur (core state machine drives the firewall). *Evidence:* host-side pcap during forced drop shows zero leaked packets (§04, §11.4.107 liveness).

AC8 — No-logging as code. No durable connection/traffic/packet table exists; the CI schema-lint fails the build if one appears. *Evidence:* `schemalint` green + paired §1.1 mutation (`connections` table → FAIL) [02 §2.4].

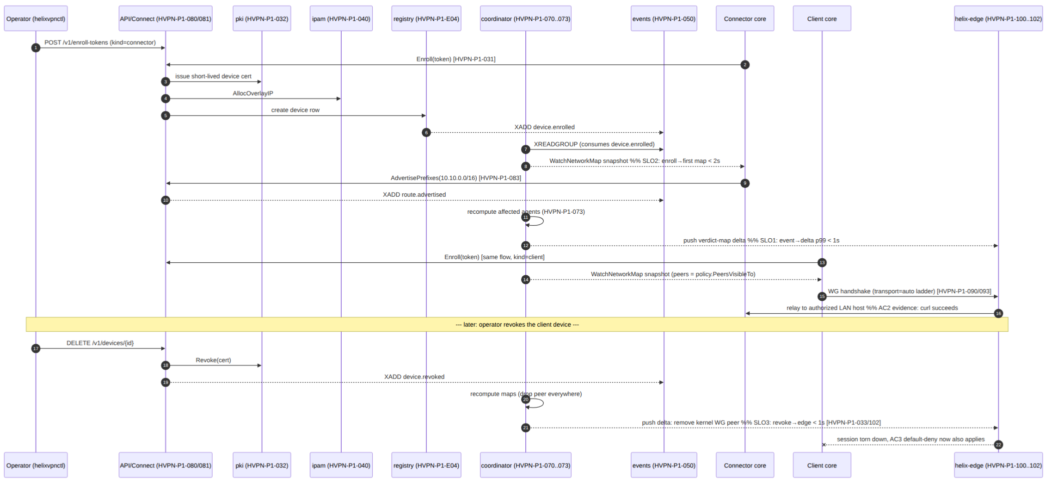
AC9 — Three apps drive it. Access (iOS/Android/Linux), Connector daemon, and Console (web) each exercise AC1–AC7 through generated clients (no hand-written API code). *Evidence:* per-app UX recordings (§11.4.159).

SLOs (measured, alert-on-breach)

ID	SLO	Target	Item that wires the metric
SLO1	event → delta-on-wire	p99 **< 1 s**	HVPN-P1-073
SLO2	enroll → first NetworkMap	**< 2 s**	HVPN-P1-031
SLO3	revoke → edge enforcement	**< 1 s**	HVPN-P1-033
SLO4	coordinator memory @ 10k streams	bounded, no leak / 24 h soak	HVPN-P1-074

2.1 Enroll → reach → revoke (the AC1/AC2/AC6/SLO2/SLO3 evidence sequence)

The single sequence that produces the captured evidence for AC1 (self-host), AC2 (authorized reach), AC6 (revoke <1s), SLO2 (enroll→map<2s) and SLO3 (revoke→edge<1s) — restated from the architecture's ASCII sketch `HelixVPN-Architecture-Refined.md` §8 as a precise Mermaid sequence diagram, annotated with the implementing work-items:



Implementing items: HVPN-P1-031/032/040/070..073/080..083/090/093/100..102/033. Proving items: HVPN-P1-140 (E2E rig), HVPN-P1-141 (SLO histograms), HVPN-P1-144 (Challenge bank + vision-verified recording).

2.2 Phase-gate-failure / rollback protocol

If the Phase-1 exit gate (all 9 ACs + 4 SLOs, §21) is **not** met on the planned schedule, the response is scoped by *which* criterion is red, never a blanket timeline extension (§11.4.42 iteration discipline):

Failure class	Response	Cross-reference
A single AC/SLO red, root cause isolated to one epic (e.g. AC5 SLO1 breach in the coordinator fan-out)	Fix-forward inside the owning epic; re-run **only** the risk-ordered set (§11.4.132) touching that epic + the full DoD sweep before re-declaring green — never spot-validate just the touched item (§11.4.130).	HVPN-P1-145 (release-gate sweep)
A **Phase-0 gate** (G1–G6, §5) regresses under Phase-1 load (e.g. G3 iOS memory headroom erodes once the reconciler + kill-switch are added)	Re-open the corresponding decision **D2/D1/D5** per `v00-meta/decision-register.md` §6 reversal criteria — this is a re-open, not a Phase-1 bug, and follows the same reopen-attribution discipline (§11.4.34) as any other regression.	`v00-meta/decision-register.md` §6

Failure class	Response	Cross-reference
Scope genuinely cannot close in the planned window (e.g. E12 platform shim effort blew past estimate)	Defer the **lowest-priority** DoD-adjacent item (never a numbered AC/SLO itself — those are the fixed exit bar) to a tracked Phase-1.1 follow-up; the release does not tag until all 9 ACs + 4 SLOs are green (§11.4.40 full-suite retest is the gate, not a target date).	§22 effort roll-up (identifies the lowest-priority non-critical-path item to defer)
A **huge-blocker** discovered during Phase-1 release validation (core capability broken, regression reaching multiple epics)	Execute the full §11.4.129 huge-blocker protocol: STOP all testing → fix all discovered issues → process the last recorded session's evidence → author new tests of every required type → rebuild → reflash → **restart** (not resume) full validation from the last tag.	§11.4.129

No phase-gate failure is silently absorbed into "ship anyway" — the DoD (§2) is the fixed floor; only the *path* to green (fix-forward / decision-reopen / scope-defer / huge-blocker-restart) is chosen based on the failure class.

---

3. Workable-item schema (§11.4.93 DB-ready)

Every leaf in §§5–20 maps 1:1 to a row in the git-tracked SQLite single source of truth (`docs/.workable_items.db`, §11.4.93/.95). Loader contract: a deterministic Go/SQL importer parses this Markdown's leaf blocks and upserts by `atm_id` (here `HVPN-P1-NNN[.k]`). The DDL the rows must satisfy:

```
-- docs/.workable_items.db — Phase-1 WBS projection (subset; full
       schema in 02/§11.4.93)
CREATE TABLE IF NOT EXISTS items (
  atm_id        TEXT PRIMARY KEY,                     -- 'HVPN-
       P1-021'
  parent_id     TEXT REFERENCES items(atm_id),        -- epic or
       parent task ('HVPN-P1-E02')
  kind          TEXT NOT NULL CHECK (kind IN
       ('epic','task','subtask')),
  title         TEXT NOT NULL CHECK (length(title) >= 6),
```

```
  description   TEXT NOT NULL CHECK (length(description) >= 40),
        -- §11.4.91/.148
  status        TEXT NOT NULL DEFAULT 'Queued'
                CHECK (status IN ('Queued','In progress','Ready
        for testing',
                                  'In
        testing','Reopened','Operator-blocked',
                                  'Obsolete (→
        Fixed.md)','Completed (→ Fixed.md)')),
  type          TEXT NOT NULL DEFAULT 'Task' CHECK (type IN
        ('Bug','Feature','Task')),
  severity      TEXT NOT NULL DEFAULT 'normal',
  epic          TEXT NOT NULL,                     -- 'E02'
  module        TEXT NOT NULL,                     --
        'store','coordinator','helix-core',...
  deps          TEXT NOT NULL DEFAULT '[]',        -- JSON array
        of atm_ids
  deliverable   TEXT NOT NULL,
  acceptance    TEXT NOT NULL,
  effort_days   INTEGER NOT NULL,
  test_types    TEXT NOT NULL,                     -- JSON:
        ["UNIT","INT","SEC"]  (§11.4.169)
  dod_refs      TEXT NOT NULL DEFAULT '[]',        -- JSON:
        ["AC3","SLO1"]
  source_refs   TEXT NOT NULL DEFAULT '[]',        -- JSON:
        ["04_P1 §7","02 §7.2"]
  created_at    TEXT NOT NULL DEFAULT (datetime('now')),
  modified_at   TEXT NOT NULL DEFAULT (datetime('now'))
);
-- per-item test diary (§11.4.149) — PASS row REQUIRES an evidence
        path:
CREATE TABLE IF NOT EXISTS test_diary (
  id           INTEGER PRIMARY KEY AUTOINCREMENT,
  atm_id       TEXT NOT NULL REFERENCES items(atm_id),
  date_time    TEXT NOT NULL,
  tested_by    TEXT NOT NULL CHECK (tested_by IN
        ('User','Operator','AI-agent','HelixQA')),
  result       TEXT NOT NULL CHECK (result IN
        ('PASS','FAIL','SKIP')),
  test_type    TEXT NOT NULL,                      -- one of
        §11.4.169 vocab
  evidence_path TEXT,                              -- qa-results/
        <run-id>/...
  observations TEXT NOT NULL,
  CHECK (result <> 'PASS' OR (evidence_path IS NOT NULL AND
        length(evidence_path) > 0))
);
```

Example row (one leaf) so the projection is unambiguous:

```
INSERT INTO items
        (atm_id,parent_id,kind,title,description,epic,module,deps,

        deliverable,acceptance,effort_days,test_types,dod_refs,source_refs)
VALUES (
```

```
'HVPN-P1-022','HVPN-P1-E02','task',
'FORCE RLS + tenant_isolation on all 10 tenant tables',
'Enable row-level security with a tenant_isolation USING/WITH
       CHECK policy on every tenant-scoped table and abort
       startup if the app role can bypass RLS.',
'E02','store','["HVPN-P1-021"]',
'migrations/0002_rls.sql; store/rls_guard.go startup check',

       'A query crafted to read tenant B from a tenant-A session
       returns ZERO rows (captured); startup aborts when role has
       rolsuper/rolbypassrls.',
4,'["UNIT","INT","SEC","CHAOS"]','["AC8","AC2","AC3"]','["02
       §2.3","04_P1 §2.2"]');
```

---

4. Epic map + dependency graph

Critical path (red): **E02 → E03 → E07 → E09 → E14**. The two Phase-0 make-or-break carries (iOS memory G3, edge language G4) are *resolved before Phase 1 starts* — Phase 1 assumes Rust core + the G4 edge decision are settled [04_P0 §0]. Epics E04/

E05/E06 fan out in parallel off E02 (disjoint scope → parallel PWUs, §11.4.58). E13 can start as soon as E07's container shape exists.

---

5. Phase 0 — entry gates (prerequisite)

> Phase 0 is **not** Phase-1 work but is the entry condition; these items track the gate verdicts that unblock the MVP. If any gate fails, the linked Phase-1 epic re-plans (e.g. G3 fail → E12.iOS adopts the §6.4 fallback ladder).

HVPN-P1-E00 — Phase-0 spike gate certification `epic · module: spike`

- **HVPN-P1-001 — G1 plain-UDP data-path gate.** `S(3) · deps: — · tests: BENCH,E2E`
  - *Desc:* Certify Rust `helix-core`+boringtun moves real traffic client→gateway→connector LAN over plain UDP at ≥80% bare-link. *Deliverable:* `bench.sh` CSV + ping log [04_P0 §4/§8]. *Acceptance:* iperf3 through-tunnel ≥80% bare link; ping to LAN host succeeds. *DoD:* AC2 (precursor).
- **HVPN-P1-002 — G2 MASQUE-through-DPI gate.** `M(5) · deps: HVPN-P1-001 · tests: BENCH,SEC,E2E`
  - *Desc:* Certify the same core tunnels over MASQUE/H3 through a UDP block at ≥50% of plain, surviving 5% loss. *Deliverable:* tshark capture (classified HTTP/3, no WG signature) + goodput CSV. *Acceptance:* works with `udp dport 51820 drop`; ≥50% plain throughput; QUIC loss-recovery beats UoT strawman. *DoD:* AC4 (precursor).
- **HVPN-P1-003 — G3 iOS NE memory gate (make-or-break).** `M(5) · deps: HVPN-P1-001 · tests: PERF,STRESS,REC`
  - *Desc:* Certify the Rust core in `NEPacketTunnelProvider` stays under the device memory ceiling with ≥30% headroom over a 30-min / 1 GB transfer, both transports. *Deliverable:* Instruments RSS report (peak+steady) [04_P0 §6]. *Acceptance:* ≥30% headroom plain **and** MASQUE; else trigger §6.4 fallback decision (§11.4.66).
- **HVPN-P1-004 — G4 edge-language decision.** `S(3) · deps: HVPN-P1-002 · tests: BENCH,PERF`
  - *Desc:* Decide Rust vs Go for the MASQUE-terminating edge on benchmark numbers (CPU/Gbps, p99, churn, MASQUE effort). *Deliverable:* §12 decision-log row + A/B CSV. *Acceptance:* recorded decision with evidence; sets E10's language (default lean Rust if within ~10–15%) [04_P0 §7]. *Decision:* D5.
- **HVPN-P1-005 — G5 FFI boundary gate.** `S(2) · deps: HVPN-P1-001 · tests: UI,E2E`
  - *Desc:* Certify flutter_rust_bridge v2 drives connect/disconnect + live status from Dart. *Deliverable:*

Flutter-Linux toggle recording. *Acceptance:* status chip transitions `Connecting→Handshaking→Connected(masque,Nms)` driven only by the core event stream. *Decision:* D2 (Rust core).

- **HVPN-P1-006 — G6 push-reconcile gate.** `XS(1) · deps: HVPN-P1-001 · tests: E2E,FA`
  - *Desc:* Certify the core converges to a static map delta with no restart (the seed of `WatchNetworkMap`). *Deliverable:* reconcile demo + map-delta log. *Acceptance:* editing `map.json` brings a new peer up with no process restart; reachability follows.

---

6. E01 — Foundation: repo, proto, codegen, CI floor

HVPN-P1-E01 — Foundation `epic · module: repo`

- **HVPN-P1-010 — Monorepo + submodule layout.** `M(4) · deps: HVPN-P1-005 · tests: UNIT,FA` *Desc:* Stand up the [SYNTHESIS §6] layout — `helix-core/ helix-edge/ helix-go/ helix-proto/ helix-ui/ shims/ deploy/` — each decoupled, reusable, snake_case (§11.4.29), with `upstreams/` + `install_upstreams` per repo (§11.4.36) and the incorporated `submodules/` (containers, helix_qa, challenges, docs_chain, security, vision_engine). *Deliverable:* repo tree + `.gitmodules` + per-repo `helix-deps.yaml` (§11.4.31). *Acceptance:* `install_upstreams` configures every declared remote; a fresh clone builds each component skeleton; **no active CI** workflows present (§11.4.156, `CM-NO-ACTIVE-CI` green). *Tests:* FA bootstrap on a throwaway clone.
- **HVPN-P1-011 — `helix-proto`: agent protobuf + buf codegen.** `M(4) · deps: HVPN-P1-010 · tests: UNIT,FA` *Desc:* Author `proto/helix/agent/v1/agent.proto` (the `Coordinator` service: `Enroll`, `WatchNetworkMap`, `AdvertisePrefixes`, `ReportStatus`, exactly as **[02 §4]** / [04_P1 §3]) and wire `buf generate` → Go + Dart + Rust stubs. *Deliverable:* generated `gen/{go,dart,rust}/...`; `buf.gen.yaml`; `buf lint`/`buf breaking` config. *Acceptance:* one source of truth → three generated clients; `buf breaking` gate fails on an incompatible change; no hand-written client code exists (§8 [04_P1]).
  - *Skeleton (the contract this WBS depends on — full text in [02 §4]):*

    ```
    service Coordinator {
      rpc Enroll(EnrollRequest) returns (EnrollResponse);
      rpc WatchNetworkMap(WatchRequest) returns (stream
            MapUpdate);
      rpc AdvertisePrefixes(AdvertiseRequest) returns
            (AdvertiseResponse);
    ```

```
  rpc ReportStatus(StatusReport) returns (StatusAck);
}
// MapUpdate.oneof{ NetworkMap snapshot | MapDelta delta |
       KeepAlive } — peers PRE-FILTERED.
```

- **HVPN-P1-012 — `helix-proto`: REST OpenAPI + Dart/TS client gen.** `S(3) · deps: HVPN-P1-011 · tests: UNIT,FA` *Desc:* Define the OpenAPI 3.1 contract for the app-facing REST surface (`/v1/enroll-tokens`, `/v1/devices`, `/v1/policies`, `/v1/networks`, `/v1/stream`) and generate Dart + TS clients. *Deliverable:* `openapi/helix.v1.yaml` + generated clients. *Acceptance:* generated client round-trips a stub server; drift between spec and handler fails the gen check.
- **HVPN-P1-013 — Local dev infra via `containers` submodule.** `S(3) · deps: HVPN-P1-010 · tests: INT,FA` *Desc:* On-demand Postgres+Redis for tests booted **only** through the `containers` submodule `pkg/boot/pkg/compose/pkg/health` (§11.4.76/.161, rootless Podman) — no ad-hoc docker. *Deliverable:* `internal/testinfra` wrapper; `make test-infra-up/down`. *Acceptance:* integration tests boot infra on demand; teardown leaves no orphan containers (§11.4.14).

---

7. E02 — Store, schema, RLS, no-log lint

HVPN-P1-E02 — Store + RLS + no-log lint (the truth floor) `epic · module: store` *Risk-first (§11.4.132): this epic is the irreversible correctness floor; it ships before any feature reads/writes data.*

- **HVPN-P1-020 — Migrations, enums, DB roles.** `M(4) · deps: HVPN-P1-013 · tests: UNIT,INT` *Desc:* Author the full DDL from **[02 §2.2]** / [04_P1 §2.1] via `goose` — `tenants users devices connectors advertised_prefixes groups group_members policies overlay_pools device_certs audit_events` + enums (`user_role`, `device_kind`); create three roles `helix_owner` (DDL), `helix_app` (non-super, non-bypass — runtime), `helix_sys` (system tasks). *Deliverable:* `migrations/0001_core.sql..`; `sqlc.yaml`. *Acceptance:* migrations apply+rollback clean on a fresh PG; `\dt` matches the spec; **no** connection/traffic table present.
- **HVPN-P1-021 — `sqlc` typed queries + `store.WithTenant`.** `M(4) · deps: HVPN-P1-020 · tests: UNIT,INT` *Desc:* Compile-time-checked queries via `sqlc`; `WithTenant(ctx,tid,fn)` runs each request inside a tx that `SET LOCAL app.tenant_id` (tx-scoped, can't leak across the pool) plus `WithSystem` for cross-tenant system tasks. *Deliverable:* `internal/store/*.go`, generated `Queries`. *Acceptance:* a query that forgets a `WHERE`

`tenant_id` still returns only the active tenant's rows (proven by INT test).

- **HVPN-P1-022 — FORCE RLS + tenant_isolation policies.** `M(4) · deps: HVPN-P1-021 · tests: UNIT,INT,SEC,CHAOS · DoD: AC8,AC2,AC3` *Desc:* `ENABLE+FORCE ROW LEVEL SECURITY` and a `tenant_isolation` `USING/WITH CHECK (tenant_id = current_setting('app.tenant_id')::uuid)` on all 10 tenant tables; a startup guard aborts if `helix_app` has `rolsuper`/`rolbypassrls`. *Deliverable:* `migrations/0002_rls.sql`; `store/rls_guard.go`. *Acceptance:* SEC test — a crafted cross-tenant read returns **zero** rows (captured); startup aborts under a super role. CHAOS — connection drop mid-tx rolls back, no leak.

```sql
ALTER TABLE devices ENABLE ROW LEVEL SECURITY;
ALTER TABLE devices FORCE ROW LEVEL SECURITY;
CREATE POLICY tenant_isolation ON devices
  USING (tenant_id = current_setting('app.tenant_id')::uuid)
  WITH CHECK (tenant_id =
        current_setting('app.tenant_id')::uuid);
-- repeat for users, connectors, advertised_prefixes, groups,
        group_members,
--           policies, overlay_pools, device_certs,
        audit_events
```

- **HVPN-P1-023 — No-logging CI schema-lint + mutation.** `S(3) · deps: HVPN-P1-020 · tests: UNIT,SEC · DoD: AC8` *Desc:* `tools/schemalint` fails the build if a durable table named/scoped like a connection log appears (`connections|flows|traffic|packets`, or any table carrying both a `src/dst` and a `bytes/ts` outside `audit_events`) [02 §2.4]. *Deliverable:* `tools/schemalint/main.go` + pre-build gate. *Acceptance:* green on the real schema; **paired §1.1 mutation** — inject a `connections(src,dst,bytes,ts)` table → lint FAILs (captured). This is the mechanical enforcement of the privacy promise.

---

8. E03 — Identity, enrollment, PKI

HVPN-P1-E03 — Identity / Enrollment / PKI `epic · module: identity,pki` *Risk-first: revoke-<1 s and key-never-leaves are security-critical (§04).*

- **HVPN-P1-030 — Dual identity: OIDC + anonymous device tokens.** `M(5) · deps: HVPN-P1-022 · tests: UNIT,INT,SEC` *Desc:* Managed mode (tenant admins log in via any OIDC IdP — Keycloak/Authentik) **and** anonymous mode (mint device enroll-tokens with no email/SSO — the Mullvad "account number, no PII" posture) [04_P1 §6.1]. *Deliverable:* `internal/identity/`; token mint/store-hashed/consume-atomic. *Acceptance:* an anonymous device enrolls with zero PII persisted; an OIDC user maps to a tenant/role; tokens are single-use and short-lived (consume is atomic under concurrency — SEC).

- **HVPN-P1-031 — `Enroll` RPC (device key never leaves).** `M(5) · deps: HVPN-P1-030,HVPN-P1-040,HVPN-P1-032 · tests: UNIT,INT,SEC,E2E · DoD: AC1,AC2; SLO2` *Desc:* Implement `Coordinator.Enroll`: verify token → IPAM-allocate overlay IP → issue short-lived device cert → persist device row (only the WG **public** key) → emit `device.enrolled` [04_P1 §6.2]. *Deliverable:* enroll handler + flow test. *Acceptance:* device generates its WG keypair locally; the control plane never receives a private key (SEC asserts absence in logs+DB); **enroll → first NetworkMap < 2 s** (SLO2, captured).
- **HVPN-P1-032 — PKI: device-cert issuance + tenant CA.** `M(4) · deps: HVPN-P1-022 · tests: UNIT,INT,SEC` *Desc:* `internal/pki` issues short-lived (24 h) mTLS device certs against a per-tenant CA; the CA root is the one true secret (KMS/offline backup) [04_P1 §6.3, §04]. *Deliverable:* `pki.Issue/Rotate/Revoke`; cert store. *Acceptance:* a cert authenticates the `WatchNetworkMap` channel; CA private key never appears in any app log (secret-leak audit, §11.4.10.A).
- **HVPN-P1-033 — Cert rotation + revoke-<1 s.** `M(4) · deps: HVPN-P1-032,HVPN-P1-070 · tests: UNIT,INT,SEC,PERF · DoD: AC6; SLO3` *Desc:* Auto-rotate before expiry over the existing channel; `device.revoke` drops the WG peer from every relevant map + enforces at the edge (kernel WG peer removed) + marks cert revoked, all within the convergence SLO. *Deliverable:* rotation scheduler + revoke path. *Acceptance:* **revoke → edge enforcement < 1 s** (SLO3, captured timing); a revoked cert is rejected immediately (SEC).

---

9. E04 — IPAM (D4)

HVPN-P1-E04 — Overlay IPAM + subnet-collision (D4)

`epic · module: ipam`

- **HVPN-P1-040 — ULA `/48` provisioning + `AllocOverlayIP`.** `M(4) · deps: HVPN-P1-022 · tests: UNIT,INT,STRESS` *Desc:* On tenant create, provision an IPv6 ULA `/48` (`fd7a:helix:<rand>::/48`, `::1` = gateway); `AllocOverlayIP` hands out the next host deterministically + concurrency-safe [02 §3.2]. **Decision D4: IPv6 ULA /48 per tenant** resolves RFC1918 collisions across N joined LANs (recommended over CGNAT-100.64 1:1 NAT) [SYNTHESIS §3 D4, 04_ARCH §3.4]. *Deliverable:* `internal/ipam/`. *Acceptance:* 1000 concurrent allocations yield no duplicate/gap (STRESS, captured); allocation is restart-stable (re-hydrates `next_host`).
- **HVPN-P1-041 — `4via6` for colliding IPv4 LANs.** `M(5) · deps: HVPN-P1-040,HVPN-P1-060 · tests: UNIT,INT,E2E` *Desc:* Map each connector site to a stable site-id and emit

Tailscale-style `4via6` routes so a client may reach two connectors that both advertise `192.168.50.0/24` without collision. *Deliverable:* site-id allocator + `4via6` route synthesis feeding `Peer.allowed_ips`. *Acceptance:* E2E — one client reaches the *same* IPv4 CIDR served by two different connectors, disambiguated by site (captured curl to each).

---

10. E05 — Event backbone (D3)

HVPN-P1-E05 — Event backbone: Redis Streams (D3) `epic · module: events`

- **HVPN-P1-050 — `events.Bus` Redis Streams impl + envelope.** `M(4) · deps: HVPN-P1-013 · tests: UNIT,INT,CHAOS` *Desc:* Bus-agnostic `events.Bus` interface over Redis Streams: `XADD` produce, `XReadGroup` durable consumer groups, `XACK`, the §5.2 envelope (`id,type,tenant_id,ts,actor,payload,trace_id`). **Decision D3: Redis Streams for MVP**, NATS JetStream is the Phase-2 swap behind the same interface [SYNTHESIS §3 D3, 04_P1 §5]. *Deliverable:* `internal/events/`. *Acceptance:* at-least-once delivery with idempotent reactions; a replayed event is harmless (INT); a consumer crash mid-handle re-delivers on restart (CHAOS).
- **HVPN-P1-051 — DLQ sweeper (`XAUTOCLAIM`) + metric.** `S(3) · deps: HVPN-P1-050 · tests: UNIT,INT,CHAOS` *Desc:* Reclaim stuck pending entries with `XAUTOCLAIM`, route exhausted-retry entries to a dead-letter, expose `helix_events_dlq_total`. *Deliverable:* sweeper goroutine + metric. *Acceptance:* a poisoned event lands in DLQ after N retries (captured), the metric increments, the live stream keeps flowing (CHAOS).
- **HVPN-P1-052 — Event taxonomy producers.** `S(3) · deps: HVPN-P1-050 · tests: UNIT,INT` *Desc:* Wire the concrete taxonomy from [04_P1 §5.3] — `device.enrolled/online/offline/revoked`, `connector.attached`, `connector.prefixes.changed`, `route.conflict.detected`, `policy.updated/compiled`, `gateway.failover` — emitted by identity/registry/policy/edge. *Deliverable:* typed event constructors. *Acceptance:* each producer emits a well-formed envelope on its trigger (INT, captured stream read).

---

11. E06 — Policy model & compiler

HVPN-P1-E06 — Policy model + compiler `epic · module: policy`

- **HVPN-P1-060 — Spec parse + group/host resolution.** `M(5) · deps: HVPN-P1-022 · tests: UNIT,INT,SEC` *Desc:* Parse the

Tailscale-ACL-flavored `policies.spec` jsonb (groups, hosts, acls, exitNodes) [04_P1 §7.1]; resolve `group:*` → device-id sets (via users + group_members) and `host` → CIDRs, cross-checking against `advertised_prefixes` (emit `route.conflict.detected` on ambiguity). *Deliverable:* `internal/policy/parse.go,resolve.go`. *Acceptance:* unknown group/host rejected; an ACL granting a revoked device rejected (SEC).

- **HVPN-P1-061 — Two-artifact compiler (AllowedIPs + verdict map).** `L(7) · deps: HVPN-P1-060 · tests: UNIT,SEC,PERF · DoD: AC2,AC3` *Desc:* Pure, deterministic, default-deny `compile(tenant,spec)→{visible,allowedIPs,exitNodes}` [04_P1 §7.2] producing **two** artefacts: coarse WG `AllowedIPs` (per-peer CIDR, data path) **and** a port-level verdict map (nftables/eBPF at the edge, since WG `AllowedIPs` is CIDR-only). Output is exactly what `coordinator.buildMap` consumes; peers are need-to-know filtered. *Deliverable:* `policy/compile.go` + golden-table tests (spec → expected visible/allowedIPs). *Acceptance:* same spec ⇒ byte-identical output (determinism property test); a contractor granted only `warehouse-cams:554,80` cannot reach port 22 (verdict-map SEC test); compile of a 1k-device tenant < 100 ms (PERF).
- **HVPN-P1-062 — Dry-run fail-closed validation + atomic activate/rollback.** `M(4) · deps: HVPN-P1-061 · tests: UNIT,INT,SEC · DoD: AC5` *Desc:* `policy.update` runs the compiler in dry-run first; reject on unknown ref, a `dst` CIDR not covered by any advertised prefix, or a grant to a revoked device. Only a clean compile bumps `policies.version`, persists the compiled artefact, and emits `policy.compiled`; a prior version re-activates instantly (rollback) [04_P1 §7.3/§7.4]. *Deliverable:* validation + version activation. *Acceptance:* a bad policy never reaches the active version (fail-closed, captured); rollback restores the prior compiled set atomically (INT).

---

12. E07 — Coordinator & WatchNetworkMap

HVPN-P1-E07 — Coordinator + WatchNetworkMap `epic · module: coordinator` *Critical path; the < 1 s convergence promise lives here.*

- **HVPN-P1-070 — In-memory per-tenant topology graph.** `M(5) · deps: HVPN-P1-052,HVPN-P1-061,HVPN-P1-032 · tests: UNIT,INT,SCALE` *Desc:* Hydrate a per-tenant graph (devices, connectors, prefixes, groups, compiled policy) from Postgres on boot and keep it fresh by consuming events; the coordinator owns **no durable tables** [04_P1 §4.1]. *Deliverable:* `internal/coordinator/graph.go`. *Acceptance:* graph reconstructable from PG + events alone (proves Phase-2 stateless-coordinator seam); applying an event mutates only the affected subgraph (INT).

- **HVPN-P1-071 — `buildMap(node)` per-node map computation.** `M(4) · deps: HVPN-P1-070 · tests: UNIT,SEC · DoD: AC2,AC3` *Desc:* `buildMap` returns `NetworkMap{self,gateway,peers,dns,transport}` where `peers = policy.PeersVisibleTo(node)` (need-to-know) each with compiled `AllowedIPs`; `relayEndpoint` always routes via the gateway in MVP (hub-and-spoke, identical to the proven Phase-0 slice) [04_P1 §4.2]. *Deliverable:* `coordinator/buildmap.go`. *Acceptance:* a node never appears in another node's map unless policy grants it (SEC — the privacy invariant, captured).
- **HVPN-P1-072 — `WatchNetworkMap` stream loop.** `L(7) · deps: HVPN-P1-071,HVPN-P1-080 · tests: UNIT,INT,E2E,CHAOS · DoD: AC5` *Desc:* Implement the streaming RPC: authenticate device by mTLS cert → send snapshot (or resume from `known_version`) → delta loop on the event subscription + keepalive ticker, with a **bounded** per-stream send queue (slow-consumer protection) [04_P1 §4.3]. *Deliverable:* `coordinator/watch.go`. *Acceptance:* reconnect with `known_version=N` resumes from deltas (no full resync, captured); a slow/stalled consumer is shed without blocking others (CHAOS).

```go
func (c *Coordinator) WatchNetworkMap(ctx context.Context,
    req *connect.Request[agentv1.WatchRequest],
    stream *connect.ServerStream[agentv1.MapUpdate]) error {
    dev := authDevice(ctx)                          // mTLS device
        cert (HVPN-P1-032)
    sub := c.subscribe(dev.TenantID, dev.ID); defer sub.Close()
    _ = stream.Send(snapshotOrResume(c.buildMap(dev),
        req.Msg.KnownVersion))
    ka := time.NewTicker(20 * time.Second); defer ka.Stop()
    for {
      select {
      case <-ctx.Done(): return ctx.Err()
      case d := <-sub.C: if err :=
        stream.Send(deltaUpdate(d)); err != nil { return err }
      case <-ka.C:       _ = stream.Send(keepAlive())
      }
    }
}
```

- **HVPN-P1-073 — Minimal-affected-set fan-out + reconcile metric.** `M(5) · deps: HVPN-P1-072 · tests: UNIT,INT,PERF · DoD: AC5; SLO1` *Desc:* On each event compute the **minimal** affected open-stream set (never broadcast a full resync on a small change) and enqueue per-stream deltas; instrument `helix_reconcile_seconds` from event-arrival to delta-on-wire [04_P1 §4.4]. *Deliverable:* fan-out + histogram. *Acceptance:* **event → delta-on-wire p99 < 1 s** (SLO1, captured histogram); a 1-device policy change touches only the affected streams (INT count assertion).

- **HVPN-P1-074 — 24 h soak / 10k-stream memory bound.** `M(5) · deps: HVPN-P1-073 · tests: SCALE,STRESS,PERF · DoD: SLO4` *Desc:* Drive N=10k simulated agents holding streams while flapping policies; assert convergence SLO holds and coordinator RSS is bounded with no leak over 24 h. *Deliverable:* `test/soak/` harness + RSS time-series. *Acceptance:* SLO1 maintained under load; **coordinator memory bounded, no growth over 24 h** (SLO4, captured graph).

---

13. E08 — API surface + authz

HVPN-P1-E08 — API: Gin REST + Connect + WS/SSE + authz `epic · module: api`

- **HVPN-P1-080 — Connect handlers mounted on the Gin listener + mTLS resolver.** `M(4) · deps: HVPN-P1-011,HVPN-P1-032 · tests: UNIT,INT,SEC` *Desc:* One multiplexed server: Connect handlers (HTTP/2 native agents, downgrade for browsers) mounted alongside Gin REST; `authDevice(ctx)` resolves the device identity from its mTLS cert [04_P1 §8]. *Deliverable:* `internal/api/server.go`, `authz_device.go`. *Acceptance:* a native agent and a gRPC-Web caller both reach `Coordinator`; an unauthenticated agent RPC is rejected (SEC).
- **HVPN-P1-081 — REST routes + OpenAPI + RBAC.** `M(5) · deps: HVPN-P1-080,HVPN-P1-012 · tests: UNIT,INT,SEC` *Desc:* Gin routes (`/v1/enroll-tokens`, `/v1/devices`, `/v1/policies`, `/v1/networks`, `/v1/audit`) behind OIDC/session-or-API-token + RBAC (`admin`/`operator`/`member`), with RLS as the backstop. *Deliverable:* route handlers + RBAC middleware. *Acceptance:* a `member` token cannot mint enroll-tokens (403, SEC); RLS prevents cross-tenant reads even if RBAC is bypassed (defense-in-depth INT).
- **HVPN-P1-082 — `GET /v1/stream` live WS/SSE fan-out.** `S(3) · deps: HVPN-P1-081,HVPN-P1-052 · tests: UNIT,INT,E2E` *Desc:* Stream control-plane events (`device.online`, `route.changed`, `handshake.failing`, `policy v N active`) to the Console over WebSocket/SSE [04_P1 §8]. *Deliverable:* `api/stream.go`. *Acceptance:* a policy apply surfaces a live "policy v N active; K devices updated" event to a connected Console within the convergence window (E2E, captured).
- **HVPN-P1-083 — `AdvertisePrefixes` + `ReportStatus` RPCs.** `S(3) · deps: HVPN-P1-080,HVPN-P1-052 · tests: UNIT,INT,SEC` *Desc:* `AdvertisePrefixes` accepts connector CIDRs (also settable via Console) emitting `connector.prefixes.changed`; `ReportStatus` carries presence + current transport + rtt **only** (deliberately no bytes/flows/destinations — the no-logging invariant on the wire). *Deliverable:* both handlers. *Acceptance:* a status report

persists no traffic data (SEC asserts the message has no byte/flow fields); advertised prefixes flow into the coordinator graph (INT).

---

14. E09 — helix-core (Rust client/connector)

HVPN-P1-E09 — helix-core: Rust client/connector `epic · module: helix-core` *The shared client/connector/edge crate; the durable evolution of the Phase-0 interfaces.*

- **HVPN-P1-090 — `Transport` trait + plain-UDP + MASQUE impls.** `L(8) · deps: HVPN-P1-002 · tests: UNIT,BENCH,SEC,E2E · DoD: AC4` *Desc:* Harden the Phase-0 `Transport` trait (`send/recv/kind/effective_mtu`, unreliable datagrams) + `plain_udp` + `masque` (quinn + h3 + hand-rolled CONNECT-UDP / HTTP-Datagram framing per RFC 9298/9297/9221) into the production crate `helix-transport` [04_P0 §4.2/§5, 03 §transport]. **Decision D1: MASQUE/QUIC primary obfuscating transport** (true Mullvad parity, single Rust impl) — surfaced against the Hysteria2 alternative [SYNTHESIS §3 D1]. *Deliverable:* `crates/helix-transport/`. *Acceptance:* MASQUE flow classifies as HTTP/3 with no WG signature (tshark, SEC); ≥50% plain throughput (BENCH); same trait drives all three consumers byte-for-byte.
- **HVPN-P1-091 — `helix-wg` boringtun wrapper.** `M(5) · deps: HVPN-P1-090 · tests: UNIT,E2E,STRESS` *Desc:* Production wrapper around boringtun `Tunn` (handshake, encrypt/decrypt, timer tick, the four `WgVerdict` outputs) — pure-userspace WG identical on Linux/Android/iOS [04_P0 §4.4]. *Deliverable:* `crates/helix-wg/`. *Acceptance:* sustained 1 GB transfer with no handshake stall (STRESS); rekey at the 2-min boundary works (captured).
- **HVPN-P1-092 — Orchestrator + network-map reconciler.** `L(7) · deps: HVPN-P1-091,HVPN-P1-072 · tests: UNIT,E2E,FA,CHAOS · DoD: AC5` *Desc:* The three-loop orchestrator (`tun→wg→transport`, `transport→wg→tun`, timer) + a reconciler that consumes streamed `MapUpdate` (snapshot/delta) and converges peers/AllowedIPs/transport **without restart**; client vs connector routing posture via mode [04_P0 §4.5, 04_P1 §3]. *Deliverable:* `crates/helix-core/`. *Acceptance:* a streamed delta adds/removes a peer with no reconnect of unrelated state (FA); a mid-stream coordinator restart re-syncs from `known_version` (CHAOS).
- **HVPN-P1-093 — Auto-escalation ladder.** `M(5) · deps: HVPN-P1-090,HVPN-P1-092 · tests: UNIT,E2E,SEC · DoD: AC4` *Desc:* Walk `TransportPolicy.order` (`["plain-udp","lwo","masque-h3"]`) on handshake failure, honoring `allow_user_override`; emit transport in `ReportStatus`. *Deliverable:* ladder state machine. *Acceptance:* with plain WG blocked the client escalates to

MASQUE and stays up (E2E, captured `transport=="masque-h3"`); ladder is deterministic across N runs (§11.4.50).

- **HVPN-P1-094 — Kill-switch + DNS-leak protection.** `L(7) · deps: HVPN-P1-092 · tests: UNIT,E2E,SEC,CHAOS · DoD: AC7` *Desc:* A core state machine drives the platform firewall so that on tunnel `Down`/`Reconnecting` **no plaintext egress and no DNS leak** occur; DNS is pinned to the overlay resolver while up [04_P1 §11.2, §04]. *Deliverable:* kill-switch module + per-OS firewall hooks (called by shims). *Acceptance:* host-side pcap during a forced drop shows **zero** leaked packets and zero plaintext DNS (SEC, captured); state survives an abrupt interface flap (CHAOS).
- **HVPN-P1-095 — `helix-ffi` (flutter_rust_bridge v2) surface.** `M(4) · deps: HVPN-P1-092 · tests: UNIT,UI,FA` *Desc:* The tiny stable FFI: `start(ClientConfig)`, `stop()`, `status_stream(StreamSink<TunnelStatus>)` mirroring the core broadcast channel (`Connecting|Handshaking|Connected{transport,rtt}|Reconnecting|Down{reason}`) [04_P0 §9]. *Deliverable:* `crates/helix-ffi/` + generated Dart bindings. *Acceptance:* Dart drives connect/disconnect and renders live status from the stream alone (UI golden + FA re-run ×3).

---

15. E10 — helix-edge (data plane)

HVPN-P1-E10 — helix-edge data plane `epic · module: helix-edge` *Language fixed by Phase-0 G4 (HVPN-P1-004); default lean Rust sharing `helix-transport` [04_P0 §7].*

- **HVPN-P1-100 — MASQUE termination on `:443/udp` + masquerade.** `L(8) · deps: HVPN-P1-090,HVPN-P1-004 · tests: UNIT,BENCH,SEC,E2E · DoD: AC4` *Desc:* Terminate MASQUE/H3, extract HTTP Datagrams → WG → kernel WG; serve a believable decoy site to anything that isn't a valid CONNECT-UDP flow (native camouflage) [04_P0 §5.4, 01]. *Deliverable:* `helix-edge/` listener. *Acceptance:* a scanner hitting `:443` sees a plain website; a valid client tunnels through (SEC + E2E, captured); CPU/Gbps within the G4 budget (BENCH).
- **HVPN-P1-101 — Kernel-WG peer programming + relay (hub-and-spoke).** `M(5) · deps: HVPN-P1-100,HVPN-P1-072 · tests: UNIT,INT,E2E · DoD: AC2` *Desc:* Program kernel WG peers from the coordinator's map and relay peer↔peer traffic via the gateway (MVP hub-and-spoke; direct P2P is Phase 2) [04_P1 §4.2]. *Deliverable:* edge↔coordinator sync + WG config apply. *Acceptance:* client reaches an authorized connector LAN host through the relay (E2E, captured curl).
- **HVPN-P1-102 — Edge verdict-map enforcement (nftables/eBPF) + revoke drop.** `M(5) · deps: HVPN-P1-061,HVPN-P1-101,HVPN-P1-033 · tests: UNIT,INT,SEC,PERF · DoD: AC3,AC6; SLO3` *Desc:* Apply the policy compiler's port-level

verdict map at the edge (the port granularity WG `AllowedIPs` lacks) and drop a revoked device's sessions on `device.revoked` [04_P1 §7.2]. *Deliverable:* nftables/eBPF programmer. *Acceptance:* an unauthorized port is dropped (AC3 SEC, captured); revoke removes the kernel peer **< 1 s** (SLO3, captured); enforcement adds negligible p99 (PERF).

- **HVPN-P1-103 — Edge hardening posture.** `S(3) · deps: HVPN-P1-100 · tests: SEC,CHAOS` *Desc:* Rootless Podman, read-only rootfs, seccomp, `NET_ADMIN`-only, **no SSH** in the edge container, fail-static (Go/control never in the packet path) [SYNTHESIS §7, §04]. *Deliverable:* edge container manifest + seccomp profile. *Acceptance:* the container runs unprivileged except `NET_ADMIN`; killing the control plane does **not** drop established data-plane tunnels (fail-static CHAOS, captured).

---

16. E11 — helix-ui (Flutter) + design system

HVPN-P1-E11 — helix-ui: Flutter design system + 3 flavors `epic · module: helix-ui` *All UI under OpenDesign tokens (§11.4.162) — light+dark, no ad-hoc CSS; Riverpod state = pure function of the core status stream [03, 04_ARCH §5].*

- **HVPN-P1-110 — `helix_design` system (OpenDesign tokens, Material 3).** `M(5) · deps: HVPN-P1-010 · tests: UNIT,UI,REC` *Desc:* Brand-tokenized design system (connection-state palette, signature components: `ConnectButton`, `StatusChip`, `ExitPicker`, `ShieldIndicator`, `AdaptiveScaffold`) with light+dark variants via OpenDesign (§11.4.162), no overlapping labels. *Deliverable:* `helix-ui/packages/helix_design/`. *Acceptance:* visual-regression goldens pass for every component in both themes (UI); a window-scoped MP4 shows the palette reacting to connection state (REC, vision-verified §11.4.159).
- **HVPN-P1-111 — Shared shell `runHelixApp(flavor,…)` + Riverpod core-status binding.** `M(4) · deps: HVPN-P1-110,HVPN-P1-095 · tests: UNIT,UI,FA` *Desc:* One tree → three flavors via `runHelixApp(flavor, home, capabilities)` (Access / Connector / Console), with Riverpod providers derived purely from the `helix-ffi` status stream [03]. *Deliverable:* `helix-ui/app/`. *Acceptance:* the same widget tree renders all three flavors; UI state is a deterministic function of a replayed status stream (FA).
- **HVPN-P1-112 — Access app screens (enroll → connect → reach).** `L(7) · deps: HVPN-P1-111,HVPN-P1-012 · tests: UI,UX,E2E,REC · DoD: AC9` *Desc:* End-user flow: paste/scan enroll token → connect → ExitPicker / authorized-network list → live status. *Deliverable:* Access flavor screens. *Acceptance:* a real enroll→connect→reach-LAN journey recorded as a window-scoped MP4 with a vision verdict that the authorized

host loaded (UX/REC, §11.4.159/.143); golden tests for each screen (UI).

- **HVPN-P1-113 — Console (web) admin screens.** `L(7) · deps: HVPN-P1-111,HVPN-P1-082 · tests: UI,UX,E2E,SEC,REC · DoD: AC5,AC9` *Desc:* The Web-only flavor (no `core_ffi`): devices, enroll-tokens, policy editor (validate→activate→rollback), live event feed via `/v1/stream`. *Deliverable:* Console flavor. *Acceptance:* a policy edit in the Console reflects on devices < 1 s with a live "policy v N active" feed (E2E/REC, AC5); RBAC hides admin actions from `member` (SEC).
- **HVPN-P1-114 — Connector daemon UI/headless control.** `M(4) · deps: HVPN-P1-111,HVPN-P1-083 · tests: UI,E2E,REC · DoD: AC9` *Desc:* The Connector flavor: enroll, advertise CIDRs, show advertised prefixes + reachability/health (runs headless on a host with a thin status UI). *Deliverable:* Connector flavor + daemon mode. *Acceptance:* a connector advertises a LAN and its prefixes appear in the Console + a client's map (E2E, captured).

---

17. E12 — Platform tunnel shims (iOS/Android/Linux)

HVPN-P1-E12 — Platform `TunnelPlatform` shims `epic · module: shims` *MVP platforms only: iOS, Android, Linux (Windows/macOS = Phase 2; HarmonyOS/Aurora = Phase 3). The shim is the **only** platform-specific code [SYNTHESIS §5].*

- **HVPN-P1-120 — Linux tun shim (Rust-native).** `S(3) · deps: HVPN-P1-092 · tests: E2E,FA · DoD: AC1,AC2` *Desc:* Linux core drives the TUN directly (kernel WG or userspace); the reference platform for the netns E2E rig. *Deliverable:* `shims/linux/`. *Acceptance:* the Access Linux build runs the full enroll→connect→reach journey on the netns rig (E2E).
- **HVPN-P1-121 — Android `VpnService` + JNI shim.** `M(5) · deps: HVPN-P1-092,HVPN-P1-094 · tests: E2E,UX,REC,SEC · DoD: AC7,AC9` *Desc:* `VpnService` owns the TUN; JNI loads `helix-core.so`; the kill-switch maps to `VpnService` always-on/lockdown semantics. *Deliverable:* `shims/android/` (Kotlin+JNI). *Acceptance:* tunnel up on a real phone (REC device recording); kill-switch blocks plaintext on drop (SEC, captured).
- **HVPN-P1-122 — iOS `NEPacketTunnelProvider` shim (within G3 budget).** `L(8) · deps: HVPN-P1-003,HVPN-P1-092,HVPN-P1-094 · tests: E2E,UX,REC,PERF,SEC · DoD: AC7,AC9` *Desc:* `NEPacketTunnelProvider` (Swift) configures `NEPacketTunnelNetworkSettings` and pumps `packetFlow` ↔ the Rust core; honor the G3 memory ceiling (apply §6.4 fallbacks only if G3 demanded them) [04_P0 §6]. *Deliverable:* `shims/apple/ios/`. *Acceptance:* tunnel up on a real device within the memory

ceiling over a sustained transfer (PERF/REC, captured Instruments RSS); kill-switch + DNS-leak hold (SEC).

---

18. E13 — Deploy: helixvpnctl + Podman quadlets + containers submodule

HVPN-P1-E13 — Deploy & operations `epic · module: deploy`

- **HVPN-P1-130 — `helixvpnctl` bootstrap + ops CLI (Cobra).** `M(5) · deps: HVPN-P1-020,HVPN-P1-130_dep · tests: UNIT,FA,E2E · DoD: AC1` *Desc:* The ops CLI replacing bash installers: `init` (generates tenant, admin, CA root, overlay `/48`, PG/Redis creds, gateway WG keys, quadlets), `gateway keys`, `enroll-token`, `policy apply`, `device revoke` [04_P1 §9.1]. *Deliverable:* `cmd/helixvpnctl/`. *Acceptance:* `helixvpnctl init` on a fresh host produces a runnable deploy (FA, captured); `policy apply ./policy.jsonc` validates + activates (E2E).

```
helixvpnctl init --domain gw.example.com --data-dir /var/lib/
helixvpn
helixvpnctl gateway keys
helixvpnctl enroll-token --kind connector --site warehouse   #
prints token / QR
helixvpnctl policy apply ./policy.jsonc
# dry-run compile → activate
helixvpnctl device revoke <id>
```

- **HVPN-P1-131 — Podman quadlets (rootless, systemd-managed).** `M(4) · deps: HVPN-P1-130,HVPN-P1-103 · tests: INT,FA,SEC · DoD: AC1` *Desc:* Render the pod quadlets (`helixd`, `helix-edge`, `postgres`, `redis`) — rootless, read-only rootfs, `helixd` connects to PG as the **non-superuser** role so RLS is enforced [04_P1 §9.2, §11.4.161]. *Deliverable:* `deploy/quadlet/*.container`, `helixvpn.pod`. *Acceptance:* `systemctl --user start helixvpn-pod` brings the stack up rootless (FA); the app role cannot bypass RLS (SEC).

```
# deploy/quadlet/helixd.container  (rootless; read-only rootfs)
[Container]
Image=ghcr.io/helixdevelopment/helixd:1.0
Pod=helixvpn.pod
Environment=DATABASE_URL=postgres://helix_app@localhost/helix
        # NON-superuser → RLS enforced
Environment=REDIS_URL=redis://localhost:6379
ReadOnly=true
[Container]  # helix-edge — winner of Phase-0 G4
Image=ghcr.io/helixdevelopment/helix-edge:1.0
```

```
Pod=helixvpn.pod
AddCapability=NET_ADMIN     # :443/udp MASQUE + kernel WG; no
        SSH in this container
```

- **HVPN-P1-132 — On-demand integration infra via `containers` submodule.** `S(3) · deps: HVPN-P1-013,HVPN-P1-131 · tests: INT,FA` *Desc:* Wire the §14 QA harness to boot the full stack on demand through the `containers` submodule (§11.4.76) so integration/E2E never require a manual `podman up`. *Deliverable:* `deploy/integration/`. *Acceptance:* the E2E suite boots+tears down the stack itself (FA, captured).
- **HVPN-P1-133 — Compose dev-loop + K8s Phase-2 shape (reference).** `S(2) · deps: HVPN-P1-131 · tests: INT` *Desc:* A Docker Compose dev-loop equivalent (local only) and a stateless-coordinator Kubernetes manifest **as reference** proving the Phase-1 seams permit HA without reshaping [02 §11.2/§11.3]. *Deliverable:* `deploy/compose/dev.yml`, `deploy/k8s/helixd-deployment.yaml`. *Acceptance:* dev compose boots PG+Redis+helixd locally; the K8s manifest is lint-valid (not deployed at MVP).

---

19. E14 — QA, SLOs, DoD certification

HVPN-P1-E14 — QA · SLOs · DoD certification `epic · module: qa` *Anti-bluff capstone: this epic owns the §11.4.40 full-suite retest, the §11.4.107 recorded evidence, and the §11.4.27 100%-type coverage that turn the 8 ACs from claims into proofs.*

- **HVPN-P1-140 — netns + DPI + netem E2E rig (CI-ready).** `M(5) · deps: HVPN-P1-101,HVPN-P1-120 · tests: E2E,FA` *Desc:* Promote the Phase-0 rig to a reproducible, fully-automated harness now fed by the **real** control plane: `lanA` netns service, `nft` WG-block for AC4, `tc netem` loss/delay, `make e2e` [04_P0 §3]. *Deliverable:* `test/e2e/rig/`. *Acceptance:* `make e2e` runs the full slice unattended and re-runs `-count=3` identically (FA, §11.4.98).
- **HVPN-P1-141 — SLO instrumentation + dashboards.** `S(3) · deps: HVPN-P1-073,HVPN-P1-074,HVPN-P1-031,HVPN-P1-033 · tests: PERF,FA · DoD: SLO1-4` *Desc:* Prometheus metrics (`helix_reconcile_seconds`, enroll/revoke timers, coordinator RSS) + Grafana-as-code dashboards + alert rules at the §2 SLO targets. *Deliverable:* `deploy/observability/`. *Acceptance:* all four SLOs measured and asserted in CI thresholds (captured histograms, alert-on-breach).
- **HVPN-P1-142 — Security test suite (RLS, key-never-leaves, mTLS, kill-switch, leak audit).** `L(7) · deps: HVPN-P1-022,HVPN-P1-031,HVPN-P1-094,HVPN-P1-102 · tests: SEC,CHAOS,SCALE · DoD: AC3,AC6,AC7,AC8` *Desc:* Consolidate the security-critical assertions into one suite driven by the `security` submodule tooling (§04): cross-tenant RLS denial, private-key

absence, mTLS rejection of bad certs, kill-switch/DNS-leak pcap, revoke timing, secret-leak repo audit (§11.4.10.A). *Deliverable:* `test/security/`. *Acceptance:* every assertion green with captured evidence; paired §1.1 mutations (drop an RLS policy / weaken kill-switch) FAIL their gate.

- **HVPN-P1-143 — Stress + chaos suite (§11.4.85).** `M(5) · deps: HVPN-P1-072,HVPN-P1-050,HVPN-P1-092 · tests: STRESS,CHAOS,SCALE` *Desc:* Sustained-load + concurrency + boundary + failure-injection across the control plane (Redis kill mid-event, PG drop mid-tx, coordinator SIGKILL with open streams, disk-full) with categorized recovery [§11.4.85]. *Deliverable:* `test/stress_chaos/`. *Acceptance:* every fault recovers to a consistent state with a captured `recovery_trace.log`; no deadlock/leak under 10× concurrency.
- **HVPN-P1-144 — HelixQA Challenge bank + vision-verified recordings.** `M(5) · deps: HVPN-P1-112,HVPN-P1-113,HVPN-P1-140 · tests: CHAL,REC,UX · DoD: AC1-AC9` *Desc:* Author a `challenges/helix_qa` bank where each Challenge dispatches a real user journey and scores PASS only on captured evidence; every user-visible AC gets a window-scoped MP4 run through the §11.4.163 media-validation pipeline (vision verdict vs SPECIFY-phase patterns) [§11.4.27/.107/.158/.159/.160]. *Deliverable:* `submodules/challenges/banks/helixvpn.yaml` + `docs/qa/<run-id>/`. *Acceptance:* the 8-AC journey set scores PASS with a vision-confirmed MP4 per AC (no frozen/bluff frame, §11.4.107); a deliberately broken build scores FAIL (analyzer self-validated golden-good/bad).
- **HVPN-P1-145 — Full-suite retest + DoD certification gate.** `M(4) · deps: HVPN-P1-140,HVPN-P1-141,HVPN-P1-142,HVPN-P1-143,HVPN-P1-144 · tests: FA,E2E,CHAL · DoD: AC1-AC9,SLO1-4` *Desc:* The §11.4.40 release-gate sweep on a clean baseline (pre-build + post-build + on-device + meta-mutation + Challenge bank + no-log lint + CONTINUATION sync), asserting all 8 ACs + 4 SLOs end-to-end on a fresh §11.4.108 deploy. *Deliverable:* `scripts/release_gate.sh` + the certification report under `docs/qa/release/`. *Acceptance:* all of §21 green simultaneously on a from-zero install; any red blocks the tag.

---

20. E15 — Docs-chain, workable-items DB, release

HVPN-P1-E15 — Docs-chain, items DB, release

`epic · module: governance`

- **HVPN-P1-150 — Workable-items DB projection + docs-chain sync.** `M(4) · deps: HVPN-P1-010 · tests: UNIT,INT,FA` *Desc:* Implement the §3 loader: parse this WBS's leaves → upsert `items/test_diary` in `docs/.workable_items.db` (git-tracked, §11.4.95) and register the docs-chain context so HTML/PDF/DOCX exports + the DB stay in lockstep (§11.4.106/.93/.65/.153). *Deliverable:* `cmd/workable-items/` + `.docs_chain/contexts/wbs.yaml`. *Acceptance:* md↔db byte-identical round-trip (§11.4.93); a leaf edit re-syncs exports out-of-the-box; `verify` is the CI gate.

- **HVPN-P1-151 — Release tagging + multi-upstream publish.** `S(3) · deps: HVPN-P1-145,HVPN-P1-150 · tests: FA · DoD: AC1-AC9` *Desc:* Cut the prefixed release tag (`<HELIX_RELEASE_PREFIX>-1.0.0-mvp` from `.env` else snake_case root dir, §11.4.151) on the main repo **and** every owned submodule with the identical prefix, published to all upstreams via merge-onto-latest-main (no force-push, §11.4.113). *Deliverable:* the tag set + changelog under `docs/changelogs/`. *Acceptance:* `git tag -l '<PREFIX>-*'` enumerates the whole release across repos; the DoD certification report (HVPN-P1-145) is the tag's evidence; the tag is reachable on every remote.

---

21. DoD ↔ work-item traceability matrix

Each release gate must trace to at least one *implementing* item and one *proving* (test/QA) item. A gate with no green proving item cannot ship (§11.4.123 rock-solid proof).

Gate	Implementing items	Proving items
AC1 self-host from zero	HVPN-P1-130, 131, 132	HVPN-P1-144, 145
AC2 authorized reach	HVPN-P1-031, 071, 101, 041	HVPN-P1-140, 144
AC3 default-deny	HVPN-P1-061, 071, 102	HVPN-P1-142, 144
AC4 transport auto-escalate	HVPN-P1-090, 093, 100	HVPN-P1-002, 140, 144
AC5 policy edit < 1 s	HVPN-P1-062, 072, 073, 113	HVPN-P1-141, 144
AC6 revoke < 1 s	HVPN-P1-033, 102	HVPN-P1-141, 142
AC7 kill-switch + DNS-leak	HVPN-P1-094, 121, 122	HVPN-P1-142, 144
AC8 no durable log	HVPN-P1-020, 023, 083	HVPN-P1-023 (mutation), 142
AC9 three apps drive it	HVPN-P1-112, 113, 114	HVPN-P1-144

Gate	Implementing items	Proving items
SLO1 event→delta p99<1s	HVPN-P1-073	HVPN-P1-141
SLO2 enroll→map <2s	HVPN-P1-031	HVPN-P1-141
SLO3 revoke→edge <1s	HVPN-P1-033, 102	HVPN-P1-141, 142
SLO4 coord mem @10k	HVPN-P1-074	HVPN-P1-141, 143

---

22. Effort roll-up & critical path

Indicative person-day totals (mid-point of the T-shirt range) for sizing, **not** a commitment — Phase-0 carries (G3/G4) are excluded as prerequisites.

Epic	Items	~Person-days
E00 Phase-0 gates (prereq)	6	19
E01 Foundation	4	14
E02 Store + RLS + lint	4	15
E03 Identity/Enroll/PKI	4	18
E04 IPAM	2	9
E05 Events	3	10
E06 Policy compiler	3	16
E07 Coordinator	5	26
E08 API + authz	4	15
E09 helix-core (Rust)	6	36
E10 helix-edge	4	21
E11 helix-ui (Flutter)	5	27
E12 Platform shims	3	16
E13 Deploy	4	14
E14 QA · SLOs · DoD	6	29
E15 Governance/release	2	7
MVP total (excl. E00)	**59**	**~273**

Critical path (longest dependency chain to AC certification): `E01(010→011) → E02(020→021→022) → E03(032) → E07(070→071→072→073) → E09(092→093/094) → E10(100→101→102) → E14(140→142→144→145) → E15(151)`. The two parallelizable fan-outs that keep wall-clock down: **E04/E05/E06** off E02 (disjoint scope → concurrent PWUs §11.4.58), and **E11 UI** off E08/E09 in parallel with E10/E12.

Audio-style "top priority first" maps here to the **security/correctness floor** (E02 RLS, E03 revoke, E09 kill-switch) leading every iteration per §11.4.132.

> Open decisions surfaced (not silently resolved), each with a recommendation and the gate that finalizes it: **D1** MASQUE/QUIC primary (rec: yes — Mullvad parity, HVPN-P1-090) · **D2** Rust core (rec: yes — G3/G5, settled) · **D3** Redis Streams MVP → NATS Phase 2 (rec: yes, HVPN-P1-050) · **D4** IPv6 ULA /48
>
> - 4via6 (rec: yes — HVPN-P1-040/041) · **D5** edge language (rec: Rust if within ~10–15%, decided by G4/HVPN-P1-004) · **D6** asymmetric per-leg transport (rec: defer to Phase 2; MVP is single-transport hub-and-spoke) · **D7** MVP ambition (rec: lean tunnel-first MVP, this WBS) [SYNTHESIS §3].

Sources

[04_P1] `HelixVPN-Phase1-MVP.md` — control-plane modular monolith, DDL+RLS, the `Coordinator` protobuf + `WatchNetworkMap`, Redis-Streams event taxonomy, policy compiler, enrollment/PKI, `helixvpnctl` + quadlets, the §11.2 8-criterion DoD + §11.3 SLOs + §11.4 no-log lint (the spine of this WBS). · [04_P0] `HelixVPN-Phase0-Spike.md` — G1–G6 entry gates, the `Transport` trait, boringtun wrapper, MASQUE/CONNECT-UDP, iOS memory harness, Go-vs-Rust edge, flutter_rust_bridge FFI, netns+DPI+netem rig. · [04_ARCH] `HelixVPN-Architecture-Refined.md` — control/data separation, ULA /48 + 4via6 (D4), no-logging, default-deny, repo layout. · [02] `02-control-plane.md` — authoritative DDL, RLS, IPAM, protobuf, events, coordinator, API, deploy manifests (this WBS references, never re-defines). · [01] `01-data-plane.md`, [03] `03-client-core-and-ui.md`, [04] `04-security-privacy-pki.md`, [00] `00-product-scope-and-principles.md`. · [05_YBO] Go+Gin+PG+Redis+Podman mandate. · [SYNTHESIS] §§1–9 — stack floor, the D1–D7 key decisions, ecosystem wiring (containers/helix_qa/challenges/docs_chain/security/vision_engine), constitution bindings (§11.4.27/.40/.58/.76/.85/.93/.106/.107/.113/.132/.151/.156/.159/.161/.162/.163/.169).